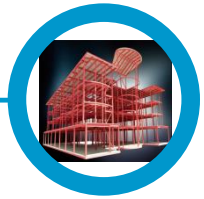




STEEL STRUCTURES/ SPACE STRUCTURES (CE-862) Fall Semester

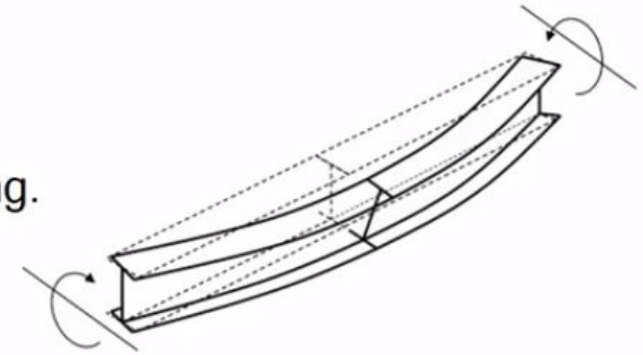
Lecture 04


Beams



Lateral Torsional Buckling

- Up until now, we haven't considered the stability issue with beams.
  - We've just assumed that the capacity will always reach M_p.
  - What if that isn't the case?
- Consider the following beam:
  - Simply supported and subjected to uniform bending.
  - The compression flange wants to buckle
  - The tension flange wants to remain straight.



- At a critical point, the beam will "kick out" and twist.
  - This phenomenon is known as lateral torsional buckling (LTB).

Activate Windows

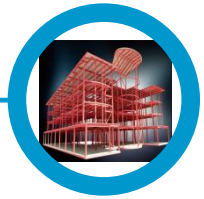
Lateral Torsional Buckling

- Examples of LTB:





Beams



3.6 Torsional stresses

When a one-dimensional element is twisted by torsional moments (see **Section 2.6**) the internal forces in the element cause torsional stresses. To see what is happening cut a slice from a circular bar that is being twisted by torsional moments.

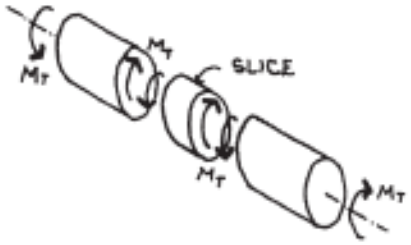


Fig. 3.65

If the circular cross-section is divided into small areas by radial and circum then each area has a force in the tangential direction to the circumference with the diagram for shear stresses—**Fig. 3.61**.



At the centre these tangential forces are zero and it is assumed that they increase linearly towards the outside of the bar. Compare this with the variation of bending stresses shown in **Fig. 3.38**.

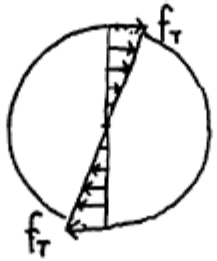
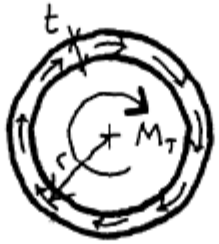


Fig. 3.67

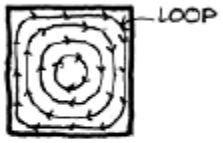
If a circular tube, with a wall thickness that is 'small' compared with its diameter, is twisted by torsional moments then it could be considered reasonable that the tangential stresses are constant across the wall of thickness **t**.



Beams



As far as torsional stresses are concerned, circular rods and tubes are special cases because their cross-section is symmetrical about **all** radial axes. The faces of slices cut elements are flat (plane) before being twisted and remain flat after the eleme twisted—this is a special case and is not true for sections of other shapes. Thi compared with the idea of plane cross-sections shown in **Fig. 3.35** for the ca beam. If a rod with an elliptical cross-section is twisted, the face of a slice that i the rod is twisted by a torsional moment will not remain flat. In this case th deform into a curved surface technically called a hyperbolic paraboloid. In othe cross-section warps and the element deforms unequally in the longitudinal dire





3.72

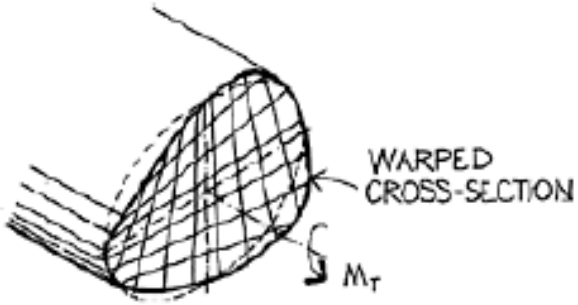


Fig. 3.70

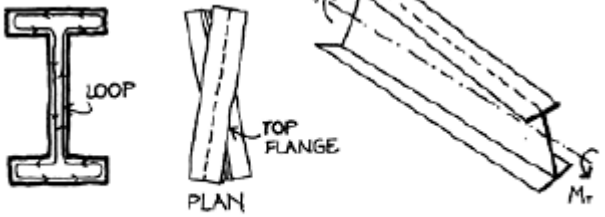
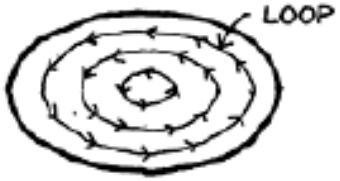
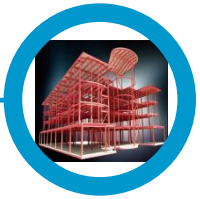


Fig. 3.73

In the case of non-circular rods, the torsional stresses can still be thought of as a series of 'loops' but in this case these loops are elliptical.



Beams



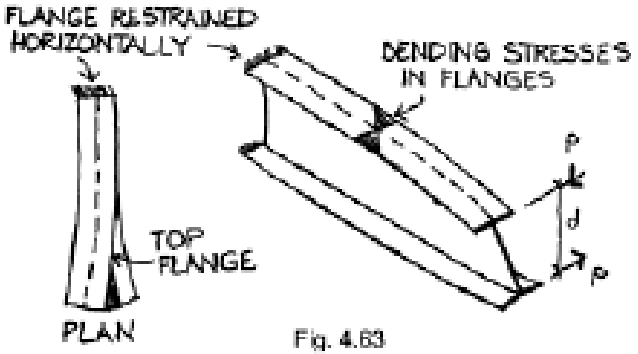
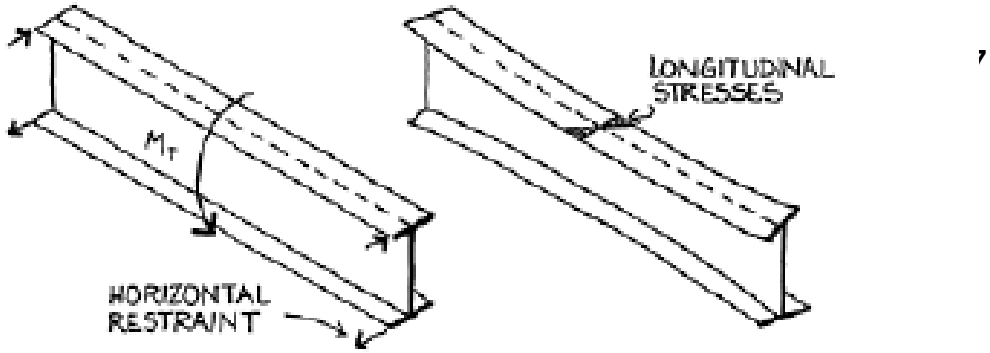
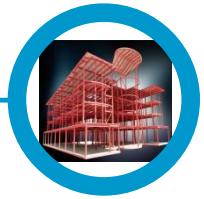


Fig. 4.63

Fig. 4.63

The torsional moment $\mathbf{M_T}$ can be thought of as a pair of point loads **P** acting on the flanges, giving the moment **P×d**. Now the torsional moment is being resisted by longitudinal stresses rather than spiral principal stresses. Although this case of horizontal restraint, or **warping restraint**, may seem an unusual one in fact it is very common in building structures. This is because a torsional moment applied between the supports of a beam gets this restraint because the flanges act as horizontal beams even if there is **no longitudinal restraint** at the support.

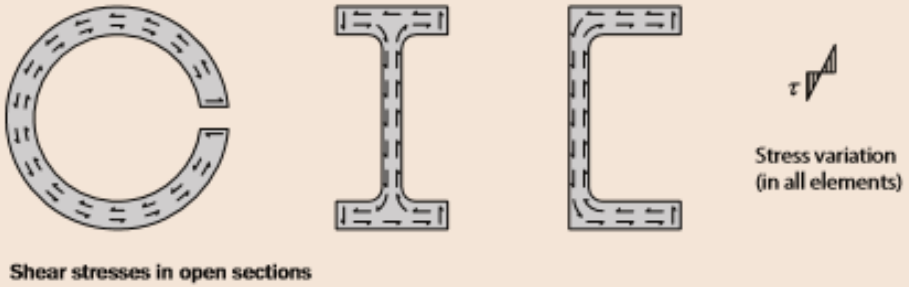




2.1 St Venant torsion

A uniform bar or beam that is subject to equal and opposite torques at each end will, if the ends are free to warp out of their planes, resist the torque at each cross section by the pattern of shear stresses shown in Figure 2.1. The total effect of the shear stresses over a cross section is equal to the torsional moment in the beam and the beam will twist about a longitudinal axis known as its shear centre (see Section 2.5 for discussion on the location of the shear centre).

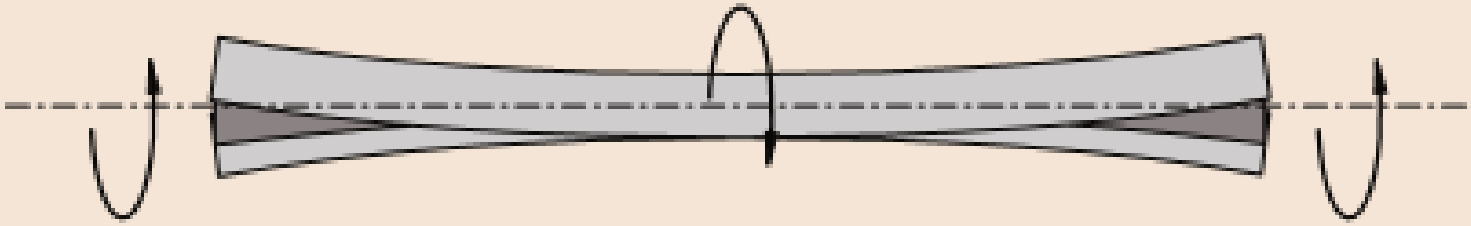
Such behaviour is sometimes referred to as 'pure torsion' but more commonly as St Venant torsion, on account of the theory developed initially by St Venant.



2.2 Warping torsion

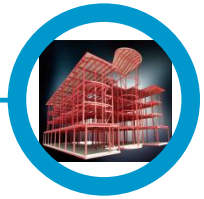
When warping of the cross sections is constrained, longitudinal stresses and additional shear stresses are developed and the torsion is partly resisted by those additional shear stresses. To illustrate the effect of warping restraint, consider a length of uniform I section with a torque applied at the middle. The displacement of the beam would then be as illustrated in Figure 2.3.

If the two halves of the beam had been separate, the left-hand half would have twisted as in Figure 2.2 and the right-hand half would have twisted in the same manner but in



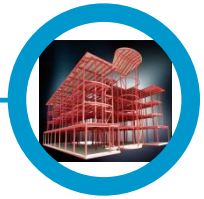
Ends free to warp

Beams



SECTION TYPE	SHAPE	ST VENANT	WARPING
Circular hollow sections	○	✓	–
Rectangular and elliptical hollow sections	□ ▯ ⬭	✓	x
Angles, Tees and cruciform sections	T L +	✓	x
Twin-flanged rolled and fabricated sections	I ⌶ ⊏	✓	✓
Thin cold-formed sections	Z Σ	x	✓

Key: ✓= significant; x = negligible; – = does not act

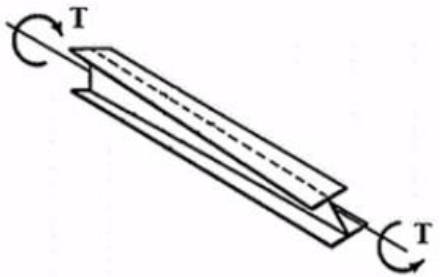


Lateral Torsional Buckling

- As stated, beams buckle by way of a torsional response.
  - Therefore, we should take some time and review torsion.

- Unlike circular shafts, I-beams respond to torsion in two fashions:

$$M_{\text{Torsion}} = M_{\text{St. Venant}} + M_{\text{Warping}}$$



  - St. Venant Torsion (or pure torsion)
  - Warping Torsion



Lateral Torsional Buckling

4. Shaft is loaded by twisting couples in planes that are perpendicular to the axis of the shaft.
5. Stresses do not exceed the proportional limit.

3-2 DERIVATION OF TORSION FORMULAS

Figure 3-1 shows two views of a solid circular shaft. If a torque T is applied at the ends of the shaft, a fiber AB on the outside surface, which is originally straight, will be twisted into a helix AC as the shaft is twisted through the angle θ. This helix is formed as follows:

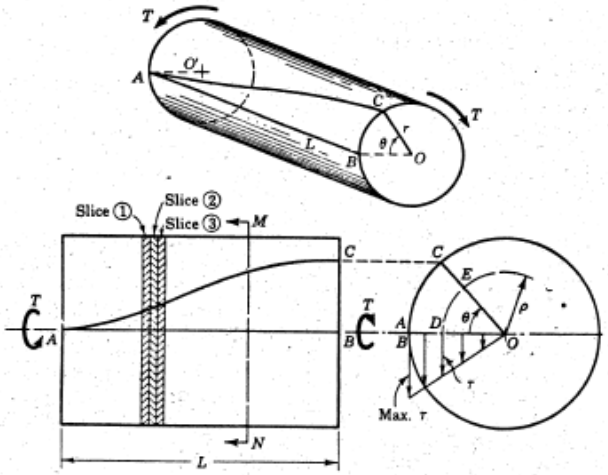


Figure 3-1 Deformation of circular shaft.

Imagine the shaft to consist of innumerable thin slices, each of which is rigid and joined to adjacent slices by elastic fibers. Slice ② will rotate past slice ① until the elastic fibers joining them are deformed enough to create a resisting torque that balances the applied torque. When this happens, slices ① and ② will act as a rigid unit and transmit the torque to slice ③; this slice will rotate enough so that the elastic fibers joining it and slice ② develop a resisting torque equal to the applied torque. This type of deformation proceeds throughout the length L of the shaft. The helix AC is the line joining the original reference line AB on these slices as they become infinitely thin. This description of the twisting action is idealized, but the resulting helix is accurately described; actually, all such slices start rotating simultaneously relative to each other as soon as the torque is applied, the angle of rotation θ becoming larger as the applied torque is increased.

Consider now any internal fiber located a radial distance ρ from the axis of the shaft. From assumption 3 in Art. 3-1, the radius of such a fiber also rotates through the angle θ, causing a total shearing deformation δ_s equal to DE. The length of this deformation is the arc of a circle whose radius is ρ and which is subtended by the angle of θ radians; the length is given by

$$\delta_s = DE = \rho\theta \qquad (a)$$

The unit deformation of this fiber is

$$\gamma = \frac{\delta_s}{L} = \frac{\rho\theta}{L} \qquad (b)$$

The shearing stress at this typical fiber is determined from Hooke's law to be

$$\tau = G\gamma = \left(\frac{G\theta}{L}\right)\rho \qquad (c)$$

Equation (c) may be called the equation of compatibility, since the stresses expressed by it are compatible with the elastic deformations. Note that each of the terms in the parentheses in this equation is a constant that does not depend on the particular internal fiber chosen for analysis; the product of these terms represents a constant. Therefore we conclude that the shearing stress at any internal fiber is determined by the product of a constant and a variable radial distance; that is, *the stress distribution along any radius varies linearly with the radial distance from the axis of the shaft.* Figure 3-1 illustrates the stress variation along the radius OB; the maximum stress occurs at the outside fiber and is denoted by Max. τ.

In line with the general procedure outlined in Art. 3-1, the shaft is divided into two segments by a cutting plane M–N. Figure 3-2 shows the free-body diagram of the left-hand portion.

A differential area of section M–N at a radial distance ρ from the axis of the shaft carries the differential resisting load $dP = \tau\, dA$. By considering an area

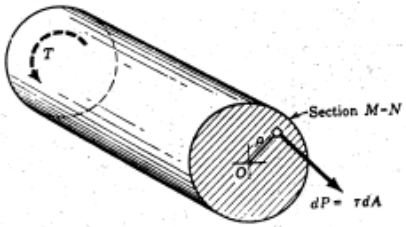
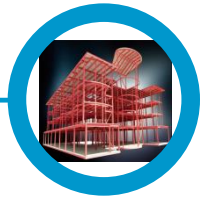


Figure 3-2 Free-body diagram of Fig. 3-1.



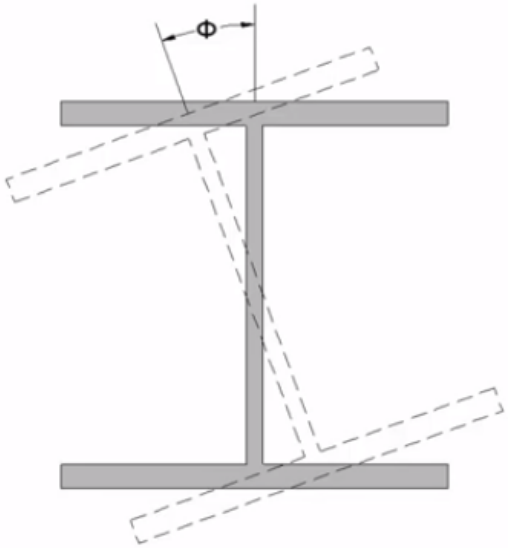
Lateral Torsional Buckling

- First on the list is pure torsion, or St. Venant torsion.
  - The same as for circular shafts.

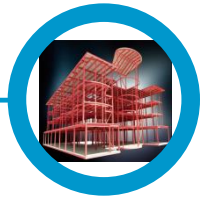
- When we apply a torsion to a section, it will respond by rotating some angle, ϕ
  - In-plane deformations.
  - Recall from mechanics:

$$\frac{M_{\text{St. Venant}}}{GJ} = \frac{d\phi}{dz}$$

$$\therefore \boxed{M_{\text{St. Venant}} = GJ\frac{d\phi}{dz}}$$

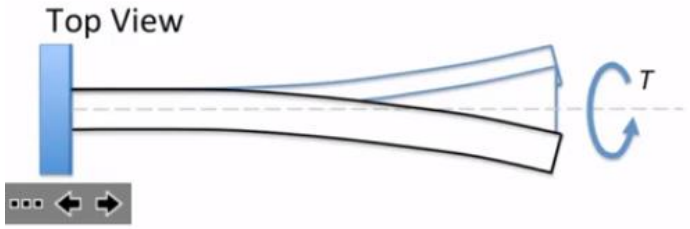


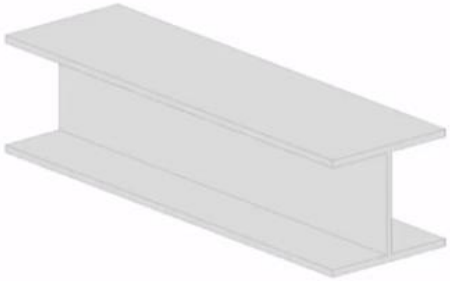
Activate Window



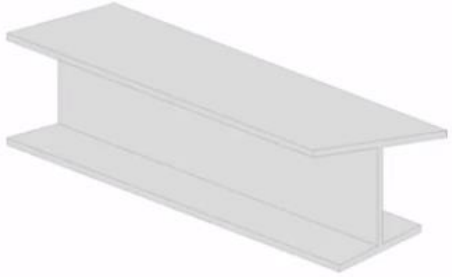
Lateral Torsional Buckling

- Warping torsion, however, takes a little bit of discussion:
  - Warping is defined the out-of-plane deformations associated with torsion.
  - Not an issue with circular cross-sections:

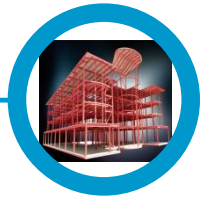




Unwarped

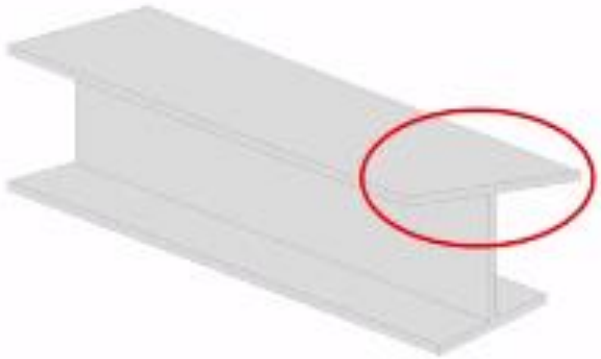


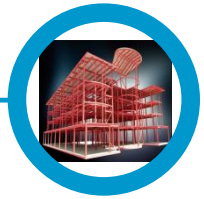
Warped



Lateral Torsional Buckling

- By observation, the majority of the warping is contained in the flanges.
- To make things simple:
  - We'll assume no warping in the web.
  - We'll focus all of our attention in the forces in the flanges.
- Ultimately we'll try and relate the shear force in the flange to the warping moment.





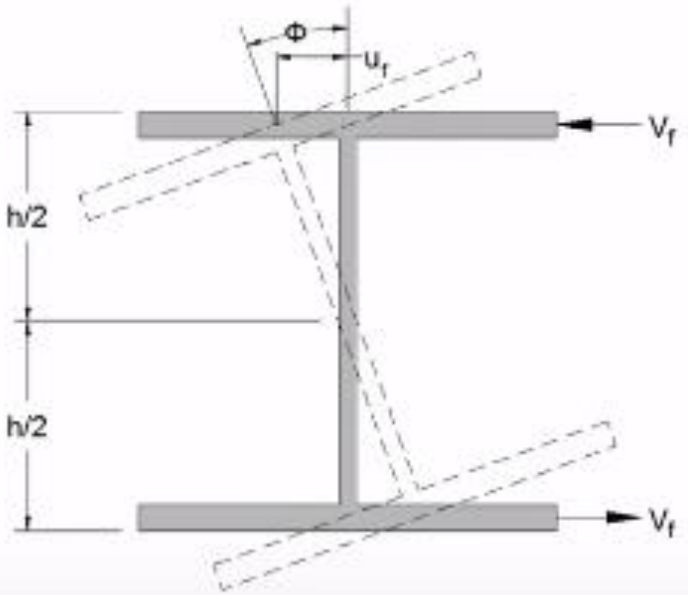
Lateral Torsional Buckling

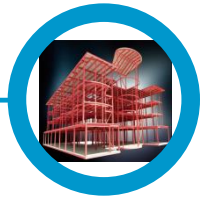
- Investigating one of the flanges, we can determine the displacement fairly easily:

$$u_f = \left(h/2\right)\phi$$

  - Also, let's take the third derivative as well (we'll use this later):

$$\frac{d^3 u_f}{dz^3} = \left(h/2\right)\frac{d^3\phi}{dz^3}$$





Lateral Torsional Buckling

- We can use this to determine the bending moment in the flange.
  - Recall from mechanics:

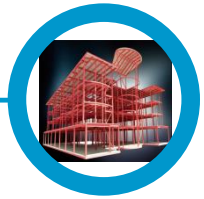
$$\frac{d^2u_f}{dz^2} = -\frac{M_f}{EI_f}$$

  - Also, recall the relationship between shear and moment:

$$\frac{dM_f}{dz} = V_f$$

- Therefore:

$$\therefore \frac{d^3u_f}{dz^3} = -\frac{V_f}{EI_f}$$



Lateral Torsional Buckling

- Setting our two expressions equal to each other:

$$\frac{d^3u_f}{dz^3} = -\frac{V_f}{EI_f} \qquad \frac{d^3u_f}{dz^3} = \left(h/2\right)\frac{d^3\phi}{dz^3}$$

$$-\frac{V_f}{EI_f} = \left(h/2\right)\frac{d^3\phi}{dz^3}$$

$$\therefore V_f = -EI_f\left(h/2\right)\frac{d^3\phi}{dz^3}$$

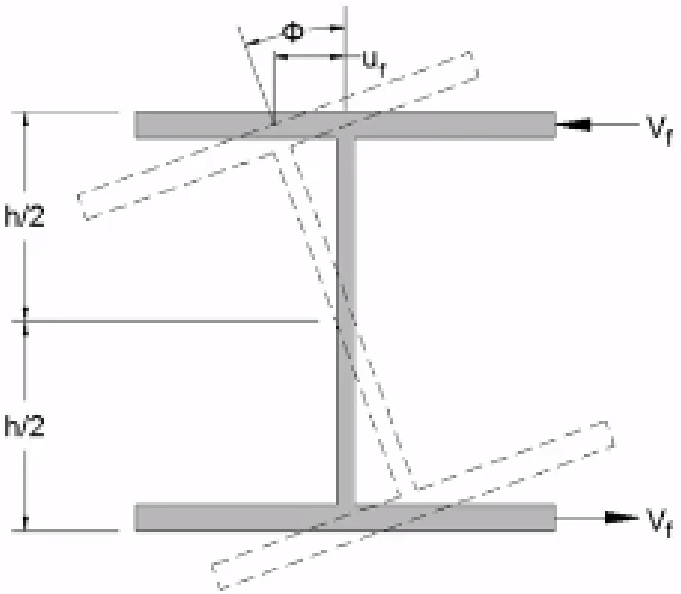
Lateral Torsional Buckling

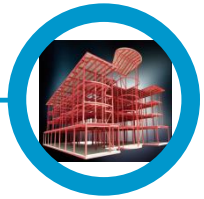
- The last thing to do is compute the moment resulting from the flange shears:

$$M_{\text{Warping}} = V_f h$$

$$M_{\text{Warping}} = \left[-EI_f \left(h/2 \right) \frac{d^3\phi}{dz^3} \right] h$$

$$\therefore M_{\text{Warping}} = -E \left(I_f h^2/2 \right) \frac{d^3\phi}{dz^3}$$





Lateral Torsional Buckling

- The term inside the parentheses is a section property of the beam:

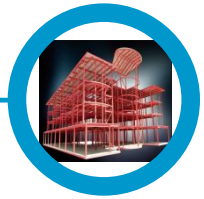
$$M_{\text{Warping}} = -E\left(I_f h^2 / 2\right)\frac{d^3\phi}{dz^3}$$

  - This is known as a warping constant:
    - Note that the majority of stiffness in the y-direction is contained in the flange.

$$C_w = \frac{I_f h^2}{2} = \frac{I_y h^2}{4}$$

- Therefore:

$$\therefore \boxed{M_{\text{Warping}} = -EC_w \frac{d^3\phi}{dz^3}}$$



Lateral Torsional Buckling

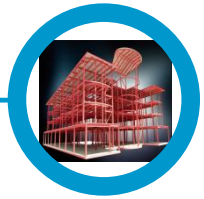
- Therefore, the differential equation for torsion in an I-section is as follows:

$$M_{\text{Torsion}} = M_{\text{St. Venant}} + M_{\text{Warping}}$$

$$M_{\text{Torsion}} = GJ\frac{d\phi}{dz} - EC_w\frac{d^3\phi}{dz^3}$$

$$\therefore \boxed{M_{\text{Torsion}} = (GJ)\phi' - (EC_w)\phi'''}$$

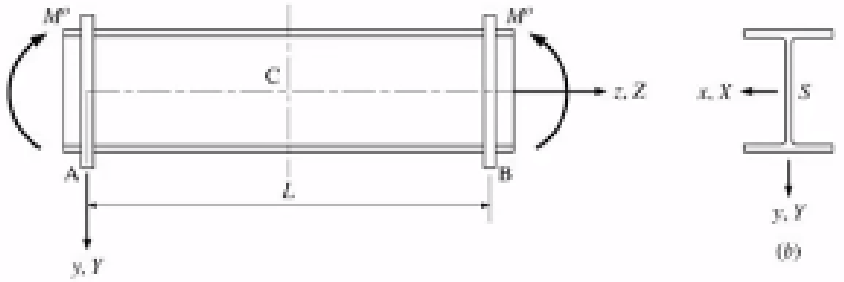
  - Now that we have an expression for the torsion in an I-section, we can derive the M_{cr} for a I-section.

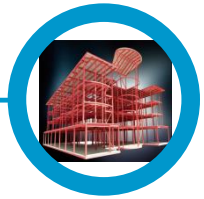


Lateral Torsional Buckling

- For this derivation, we'll consider a beam subjected to uniform moment:
  - Supports prevent warping:

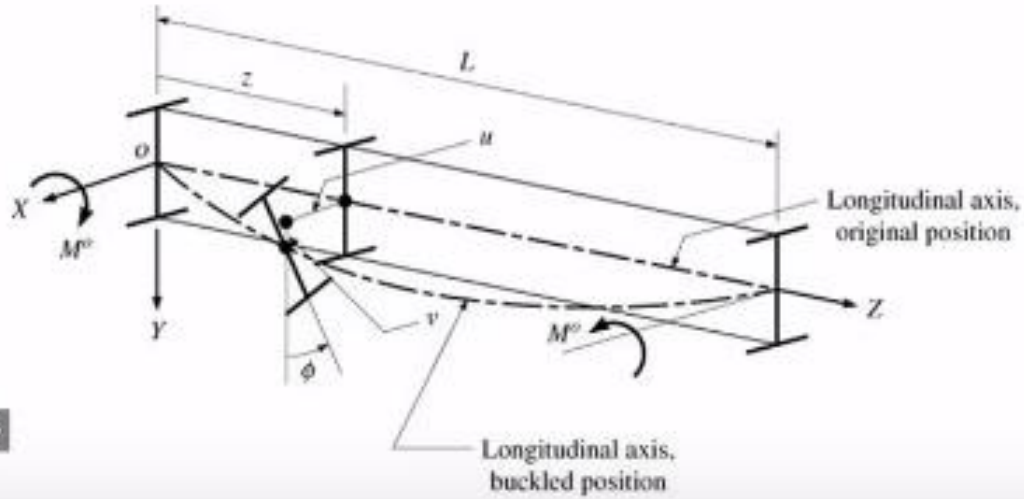
- We'll have three moments that we'll have to consider:
  - Vertical deflection (yz plane)
  - Lateral deflection (xz plane)
  - Twisting (along z-axis)





Lateral Torsional Buckling

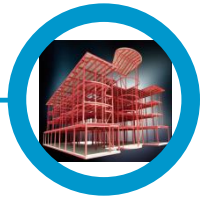
- For the yz plane (vertical deflection):



$$M_o = -EI_x \frac{d^2v}{dz^2}$$

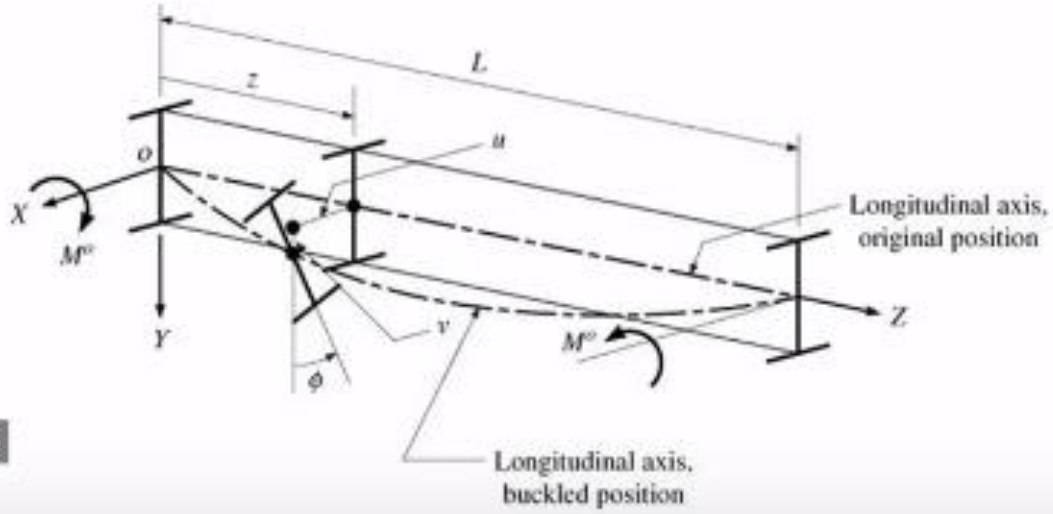
$$\therefore \boxed{M_o = -(EI_x)v''} \quad \boxed{1}$$

Note:
We won't use Eq. 1 directly. However, it serves as a reminder of general beam deflection theory.



Lateral Torsional Buckling

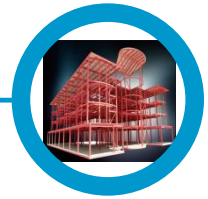
- For the xz plane (lateral deflection):



$$M_o \sin(\phi) = -EI_y \frac{d^2u}{dz^2}$$

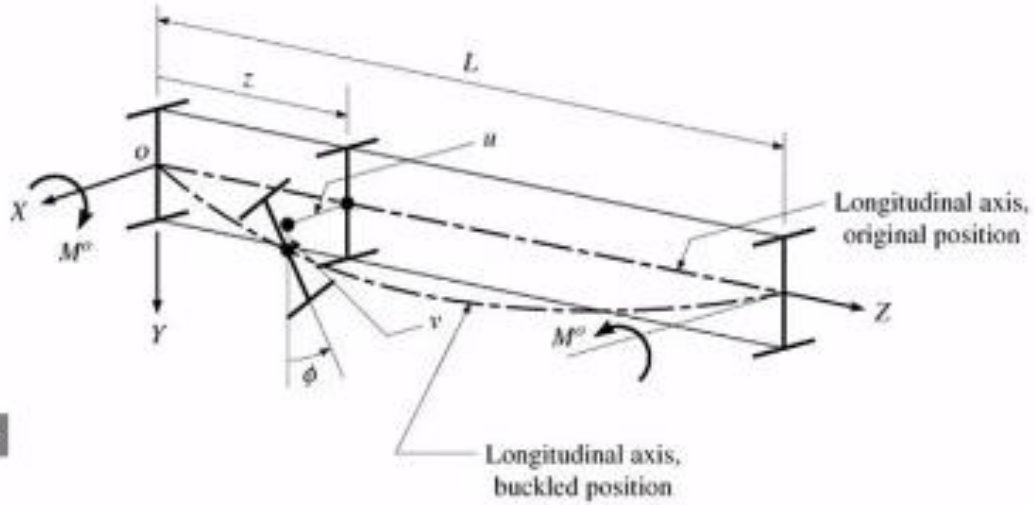
$$\sin(\phi) \approx \phi$$

$$\therefore \boxed{M_o\phi = -(EI_y)u''} \quad \boxed{2}$$



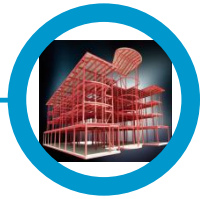
Lateral Torsional Buckling

- For the twisting moment (already derived):



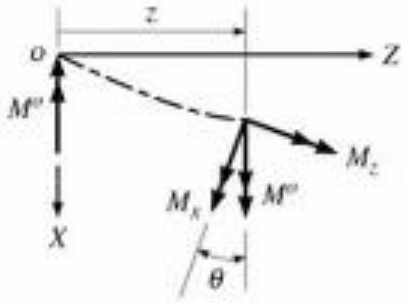
$$M_z = GJ\frac{d\phi}{dz} - EC_w\frac{d^3\phi}{dz^3}$$

$$\therefore \boxed{M_z = (GJ)\phi' - (EC_w)\phi'''} \quad \boxed{3}$$

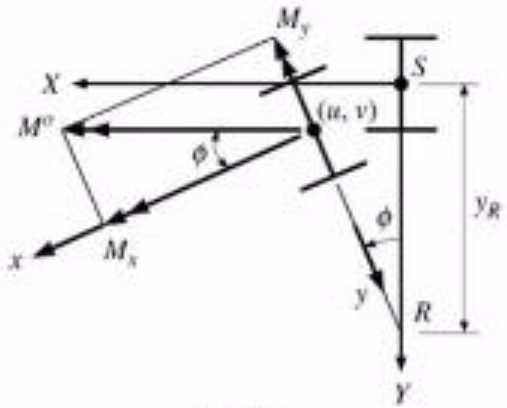


Lateral Torsional Buckling

- To combine these into a useful expression, we'll use the following:
  - The slope of the beam, θ, and the lateral deflection can be related:



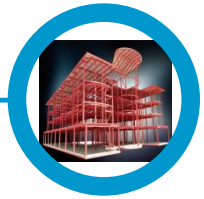
View from above



End view

$$M_z = -M_o\theta = -M_o\frac{du}{dz}$$

$$\therefore \boxed{M_z = -(M_o)u'} \quad \boxed{4}$$



Lateral Torsional Buckling

- Therefore, setting equations 3 and 4 equal to each another:

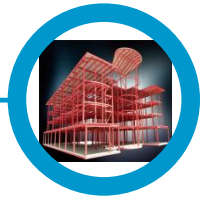
$$M_z = (GJ)\phi' - (EC_w)\phi''' \quad \boxed{3}$$

$$M_z = -(M_o)u' \quad \boxed{4}$$

$$-(M_o)u' = (GJ)\phi' - (EC_w)\phi'''$$

  - Also, if we take the derivative again:

$$-(M_o)u'' = (GJ)\phi'' - (EC_w)\phi^{IV}$$



Lateral Torsional Buckling

- The reason for the additional derivative is to use equation 2:

$$M_o\phi = -\left(EI_y\right)u'' \quad \boxed{2}$$

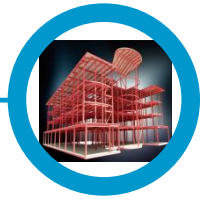
$$\therefore u'' = -\left(\frac{M_o}{EI_y}\right)\phi$$

- Plugging in:

$$-\left(M_o\right)u'' = \left(GJ\right)\phi'' - \left(EC_w\right)\phi^{IV}$$

$$\left(\frac{M_o^{\,2}}{EI_y}\right)\phi = \left(GJ\right)\phi'' - \left(EC_w\right)\phi^{IV}$$

$$\therefore \boxed{\left(EC_w\right)\phi^{IV} - \left(GJ\right)\phi'' - \left(\frac{M_o^{\,2}}{EI_y}\right)\phi = 0}$$

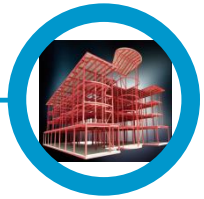


Lateral Torsional Buckling

- We now have a 4th order differential equation to solve:

$$(EC_w)\phi^{IV} - (GJ)\phi'' + \left(\frac{M_o^{\,2}}{EI_y}\right)\phi = 0$$

- We could accomplish this two ways:
  - Solve the characteristic equation and apply boundary conditions.
    - The hard way
  - Assume a solution that meets boundary conditions (in other words, deflection and moment at each end of the unbraced length equals zero).
    - The easy way



Lateral Torsional Buckling

- When we looked at columns, we found that the solution was sinusoidal.
  - Let's assume the twisting of the beam behaves in the same manner:
    - k = twist at midspan
    - L_b = unbraced length (distance between bracing points)
  - Also, we'll go ahead and take the necessary derivatives:

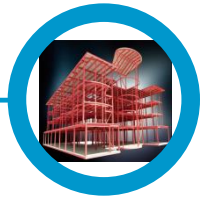
$$\phi(z) = k \sin\left(\pi z / L_b\right)$$

$$\phi'(z) = \frac{d\phi}{dz} = \left(k \frac{\pi}{L_b}\right) \cos\left(\pi z / L_b\right)$$

$$\phi''(z) = \frac{d^2\phi}{dz^2} = -\left(k \frac{\pi^2}{L_b^2}\right) \sin\left(\pi z / L_b\right)$$

$$\phi'''(z) = \frac{d^3\phi}{dz^3} = -\left(k \frac{\pi^3}{L_b^3}\right) \cos\left(\pi z / L_b\right)$$

$$\phi^{IV}(z) = \frac{d^4\phi}{dz^4} = \left(k \frac{\pi^4}{L_b^4}\right) \sin\left(\pi z / L_b\right)$$



Lateral Torsional Buckling

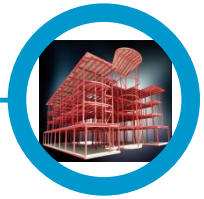
- Plugging in:

$$(EC_w)\phi^{IV} - (GJ)\phi'' + \left(\frac{M_o^{\,2}}{EI_y}\right)\phi = 0$$

$$EC_w\left[\left(k\frac{\pi^4}{L_b^{\,4}}\right)\sin\left(\pi z/L_b\right)\right] - GJ\left[\left(k\frac{\pi^2}{L_b^{\,2}}\right)\sin\left(\pi z/L_b\right)\right] + \frac{M_o^{\,2}}{EI_y}\left[k\sin\left(\pi z/L_b\right)\right] = 0$$

  - Factoring out common terms:

$$\left[EC_w\left(\frac{\pi^4}{L_b^{\,4}}\right) - GJ\left(\frac{\pi^2}{L_b^{\,2}}\right) + \frac{M_o^{\,2}}{EI_y}\right]k\sin\left(\pi z/L_b\right) = 0$$



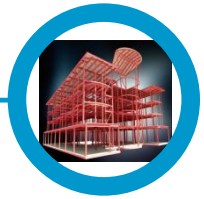
Lateral Torsional Buckling

- The only way that we can achieve a nontrivial solution is if the term inside the brackets is equal to zero:

$$\left[EC_w\left(\frac{\pi^4}{L_b^4}\right)-GJ\left(\frac{\pi^2}{L_b^2}\right)+\frac{M_o^2}{EI_y}\right]k\sin\left(\pi z/L_b\right)=0$$

$$EC_w\left(\frac{\pi^4}{L_b^4}\right)-GJ\left(\frac{\pi^2}{L_b^2}\right)+\frac{M_o^2}{EI_y}=0$$

The only thing left to do is solve for the moment.



Lateral Torsional Buckling

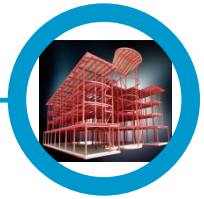
- Solving:

$$EC_w\left(\frac{\pi^4}{L_b^4}\right)-GJ\left(\frac{\pi^2}{L_b^2}\right)+\frac{M_o^2}{EI_y}=0$$

$$\frac{M_o^2}{EI_y}=GJ\left(\frac{\pi^2}{L_b^2}\right)-EC_w\left(\frac{\pi^4}{L_b^4}\right)=\frac{\pi^2}{L_b^2}\left[GJ-EC_w\left(\frac{\pi^2}{L_b^2}\right)\right]$$

$$M_o^2=\frac{\pi^2 EI_y}{L_b^2}\left[GJ-EC_w\left(\frac{\pi^2}{L_b^2}\right)\right]=\frac{\pi^2}{L_b^2}\left[EI_yGJ-I_yC_w\left(\frac{\pi^2E^2}{L_b^2}\right)\right]$$

$$\therefore\ M_o=\frac{\pi}{L_b}\sqrt{EI_yGJ-\left(\frac{\pi E}{L_b}\right)^2 I_yC_w}$$



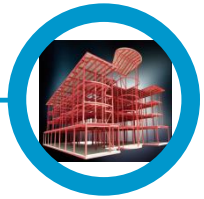
Lateral Torsional Buckling

- Therefore, the elastic buckling moment, M_{cr}, according to LTB is:

$$M_{cr} = M_o = \frac{\pi}{L_b}\sqrt{EI_yGJ - \left(\frac{\pi E}{L_b}\right)^2 I_y C_w}$$

  - To obtain F_{cr}, we simply divide M_{cr} by the section modulus, S_x:

$$F_{cr} = \frac{M_{cr}}{S_x} = \frac{\pi}{L_b S_x}\sqrt{EI_yGJ - \left(\frac{\pi E}{L_b}\right)^2 I_y C_w}$$



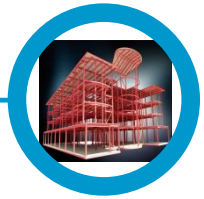
Lateral Torsional Buckling

- To obtain the equation presently in the Specifications, the following substitutions are made:
  - For steel, E = 29000 ksi, G = 11200 ksi

$$C_w = \frac{I_f h_o^2}{2} = \frac{I_y h_o^2}{4} \qquad r_{ts}^2 = \frac{\sqrt{I_y C_w}}{S_x} \qquad h_o = d - t_f \qquad \frac{2G}{\pi^2 E} \approx 0.078$$

- After a little bit of algebra:

$$F_{cr} = \frac{\pi}{L_b S_x}\sqrt{EI_y GJ - \left(\frac{\pi E}{L_b}\right)^2 I_y C_w} \quad \rightarrow \quad \boxed{F_{cr} = \frac{\pi^2 E}{\left(L_b / r_{ts}\right)^2}\sqrt{1 + 0.078\frac{J}{S_x h_o}\left(L_b / r_{ts}\right)^2}}$$



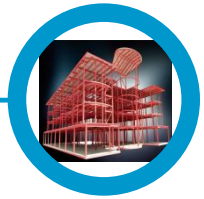
Lateral Torsional Buckling

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Lateral Torsional Buckling

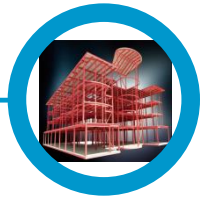
- So, the equation that we just derived is:

$$F_{cr} = \frac{\pi^2 E}{\left(L_b / r_{ts}\right)^2} \sqrt{1 + 0.078 \frac{J}{S_x h_o} \left(L_b / r_{ts}\right)^2}$$

  - If we open the Specifications, we'll find:

$$F_{cr} = \frac{C_b \pi^2 E}{\left(L_b / r_{ts}\right)^2} \sqrt{1 + 0.078 \frac{Jc}{S_x h_o} \left(L_b / r_{ts}\right)^2}$$

- The "c" factor adjusts the equation for C-shapes.
  - What about the "C_b" factor?

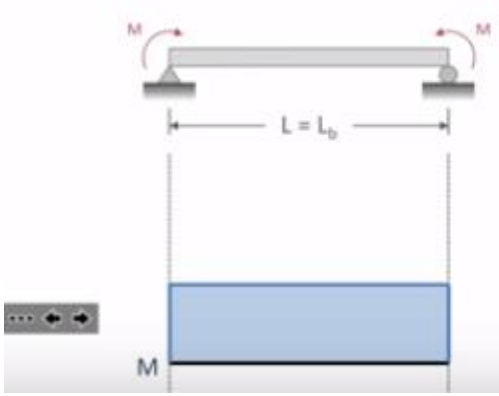


Lateral Torsional Buckling

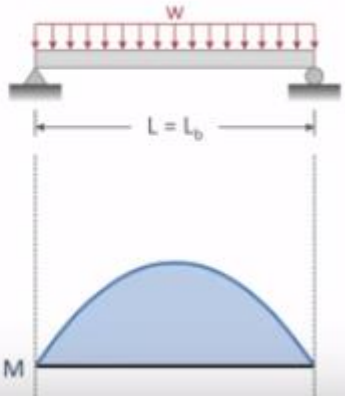
- When we performed our derivation, we assumed that bending moment was a constant.
  - This allowed us to factor the moment term out of the differential equation.
  - If the moment was a variable, that would have made the derivation really difficult!

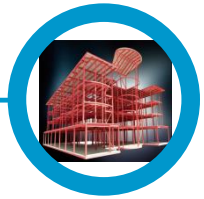
- C_b is called the moment gradient modifier
  - It accounts for non-uniform moments in an unbraced length.
  - Something we deal with quite a bit!

- Uniform bending:



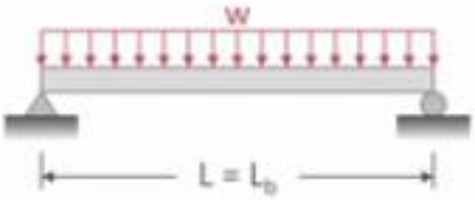
- Example of non-uniform bending:



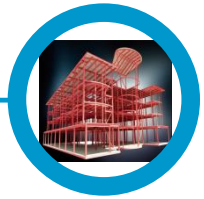


Lateral Torsional Buckling

- When we assume that moment is constant in a beam, we're saying that every point in the beam must be expected to reach M_{cr}.
  - However, if we're analyzing a beam like this, we only need to develop M_{cr} at midspan.



…e problem, however, is if we try and derive an expression for every case of non-uniform moment, the differential equation gets much more complex if moment varies.



Lateral Torsional Buckling

- Luckily, this problem has been addressed by researchers.
  - Many expressions for C_b have been derived over the years.
  - However, the one in the AISC Specifications is pretty straightforward.

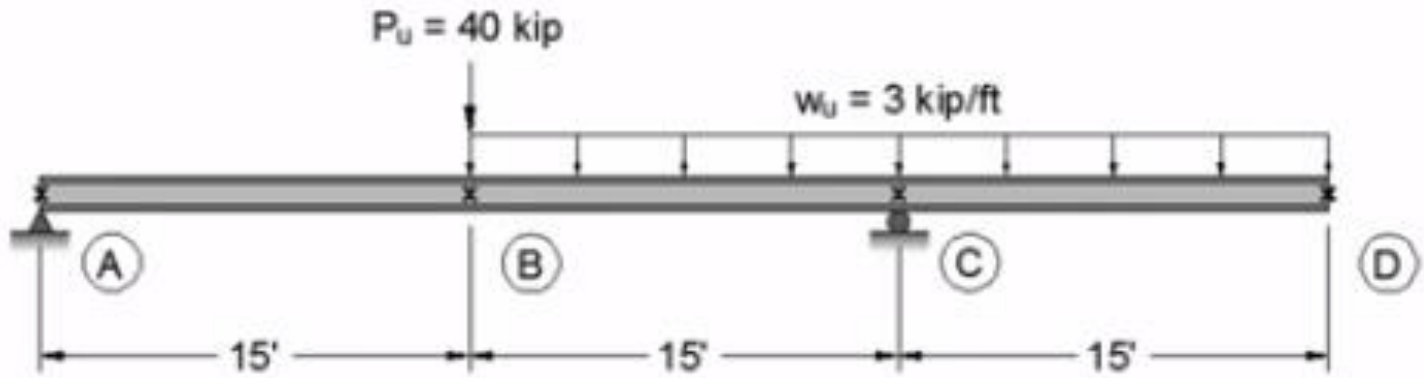
$$C_b = \frac{12.5M_{max}}{2.5M_{max} + 3M_A + 4M_B + 3M_C}$$

Eqn. F1-1, Pg. 16.1-46

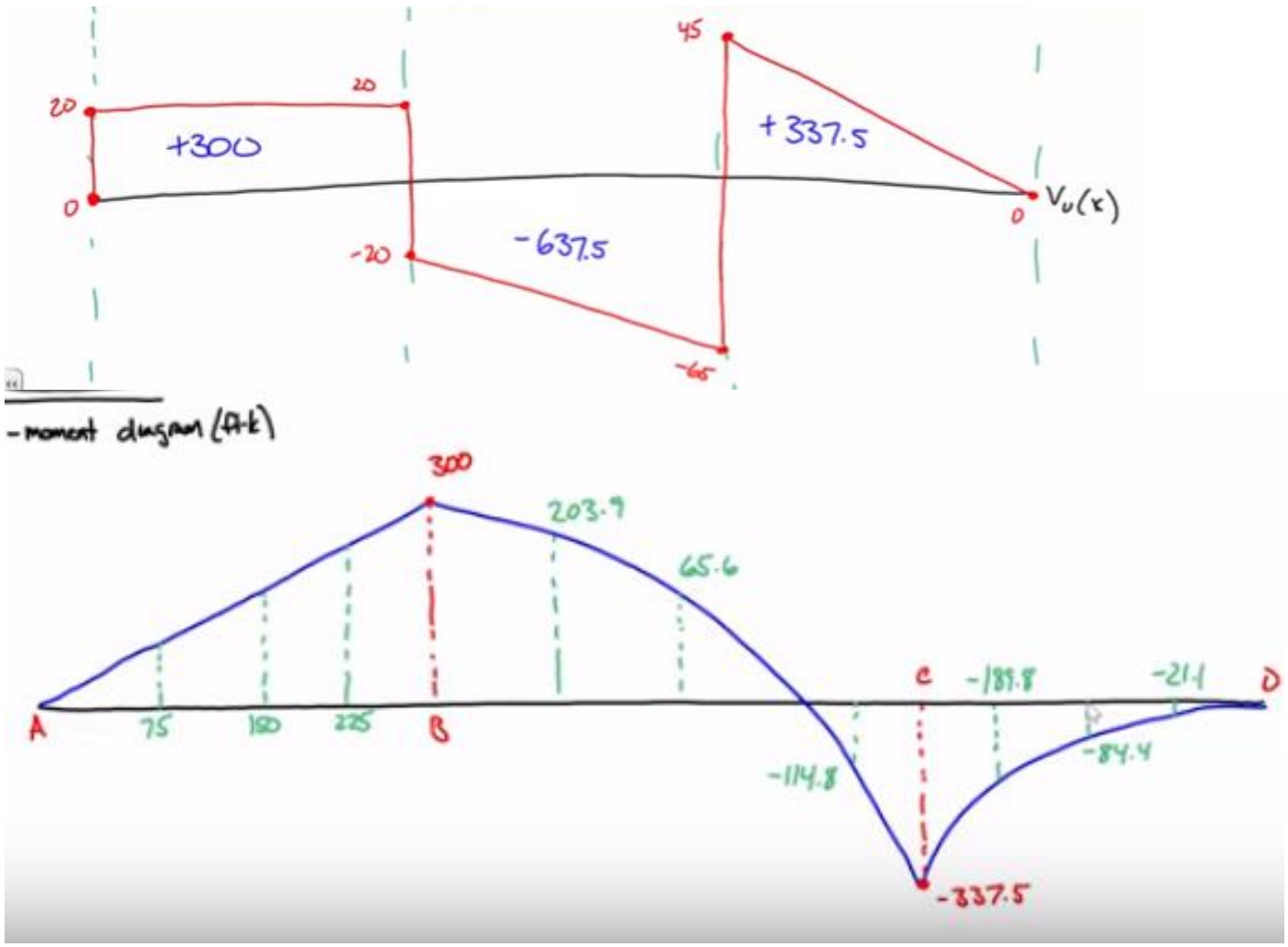
  - Within a given unbraced length:
    - M_A, M_B, M_C = quarter point moments
    - M_{max} = maximum moment
  - All moments are taken as absolute values!

Lateral Torsional Buckling

- Compute C_b values for each unbraced segment on the beam shown:
  - Lateral bracing at A, B, C, and D
  - All loads have been factored.



Lateral Torsional Buckling



Lateral Torsional Buckling

AB

$M_A = 75^{k}$

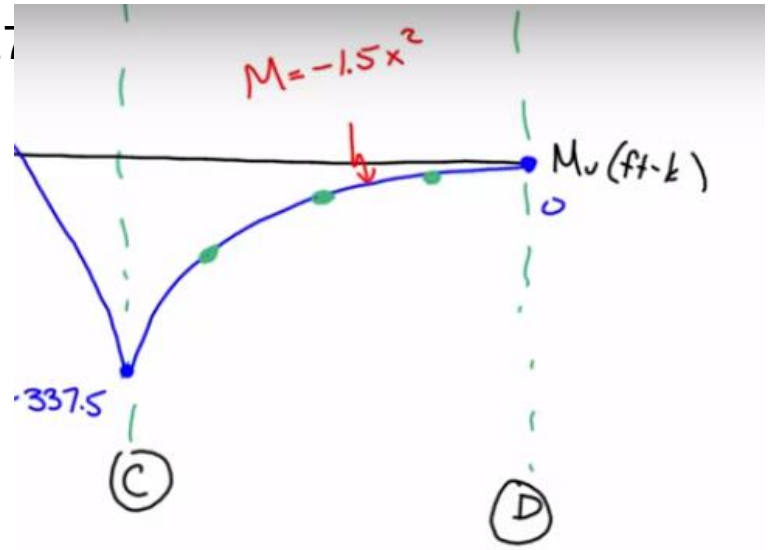
$M_B = 150^{k}$

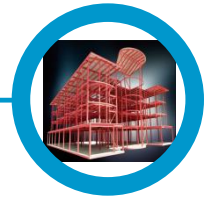
$M_C = 225^{k}$

$M_{max} = 300^{k}$

$$\therefore C_b = \frac{12.5 M_{max}}{2.5 M_{max} + 3M_A + 4M_B + 3M_C} = 1.67$$

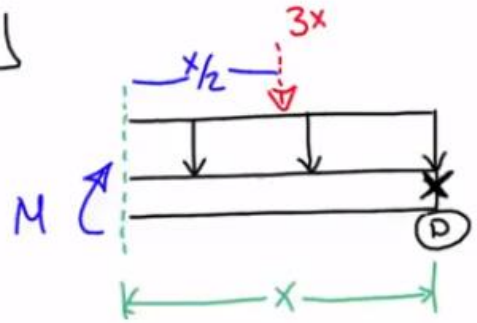
Page-16.1-17





Lateral Torsional Buckling

CD



$\Sigma M_{cut} = 0$

↺	↻
M	
$3x(x/2)$	

$M + \frac{3}{2}x^2 = 0$

$\therefore M = -\frac{3}{2}x^2$

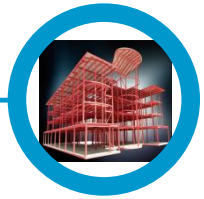
$M_A = +21.1^{lk}$

$M_B = +84.4^{lk}$

$M_C = +189.8^{lk}$

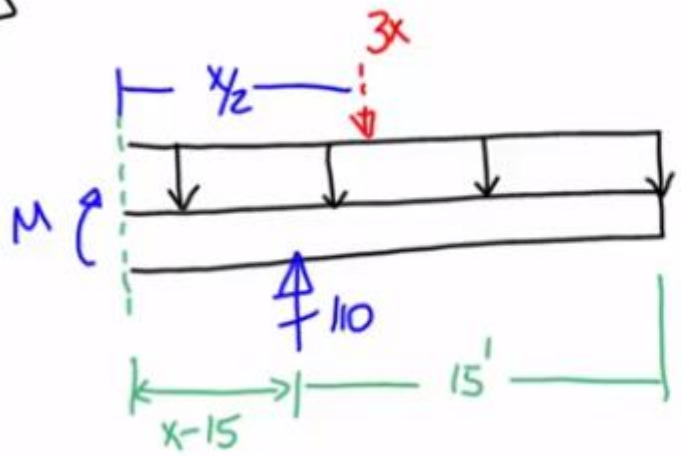
$M_{max} = +337.5^{lk}$

$C_b = 2.33$



Lateral Torsional Buckling

BC



$$M = -\frac{3}{2}x^2 + 110(x-15)$$

↱	↰
M 3x(x/2)	110(x-15)

@ x = 15' + 3.75'

@ x = 15' + 7.5'

@ x = 15' + 11.25

$M_A = +114.8^{k}$

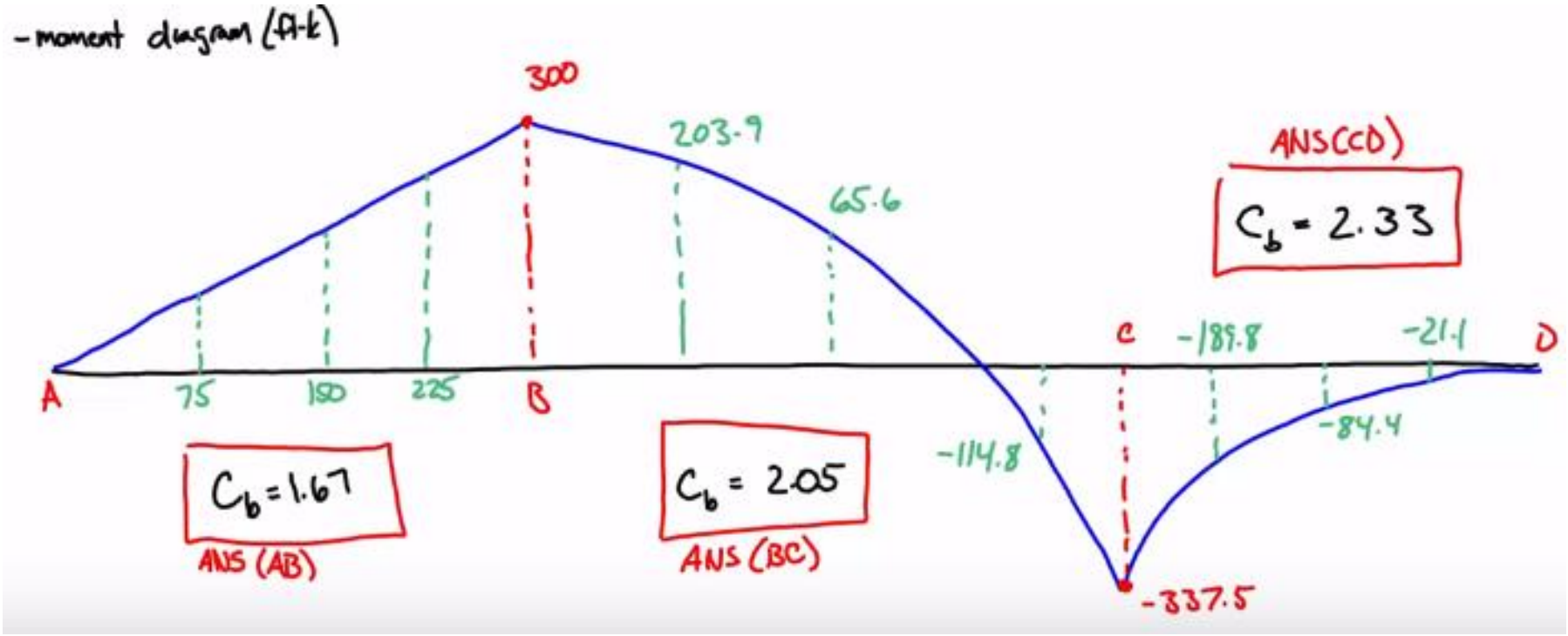
$M_B = +65.6^{k}$

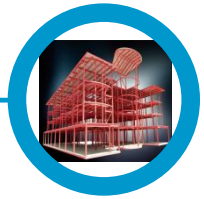
$M_C = +203.9^{k}$

$M_{max} = +337.5^{k}$

$C_b = 2.05$

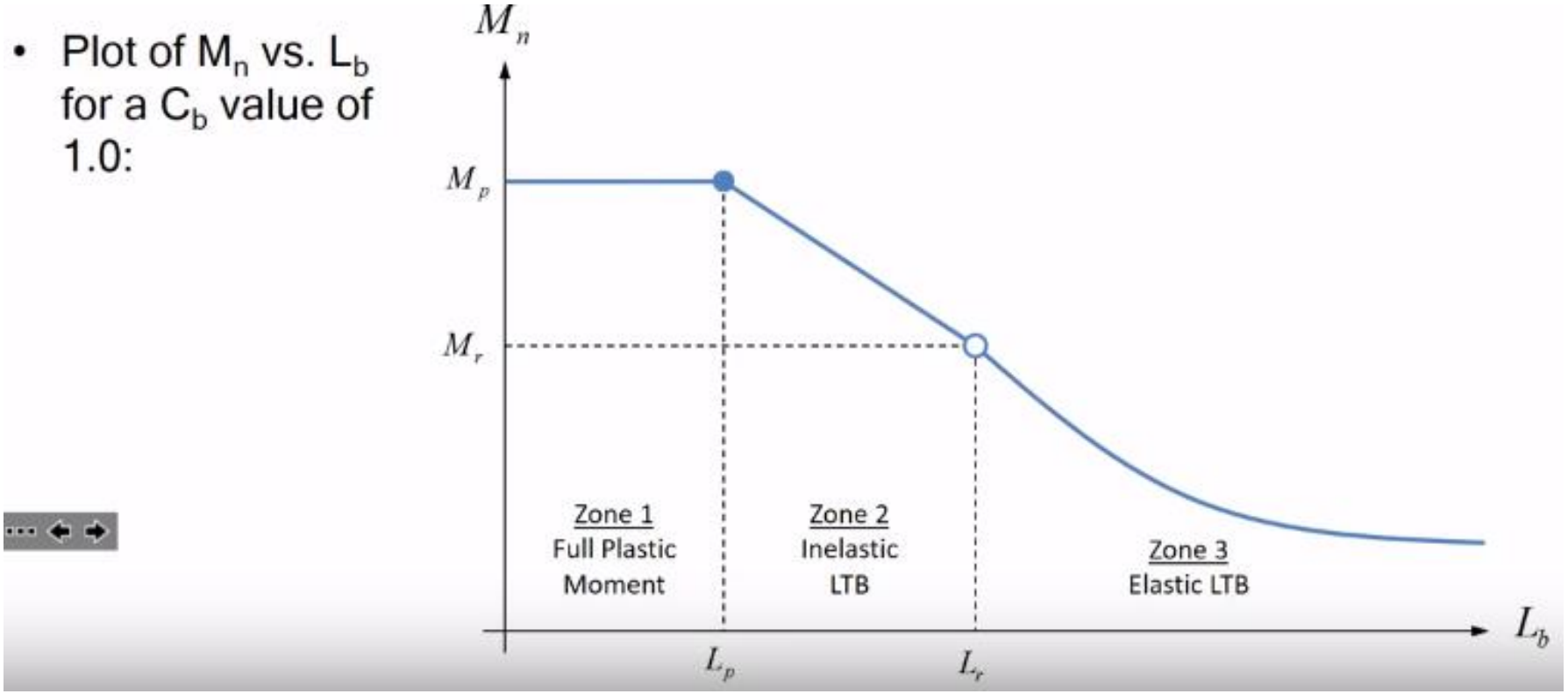
Lateral Torsional Buckling

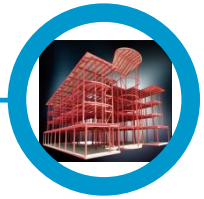




Discretely braced beams

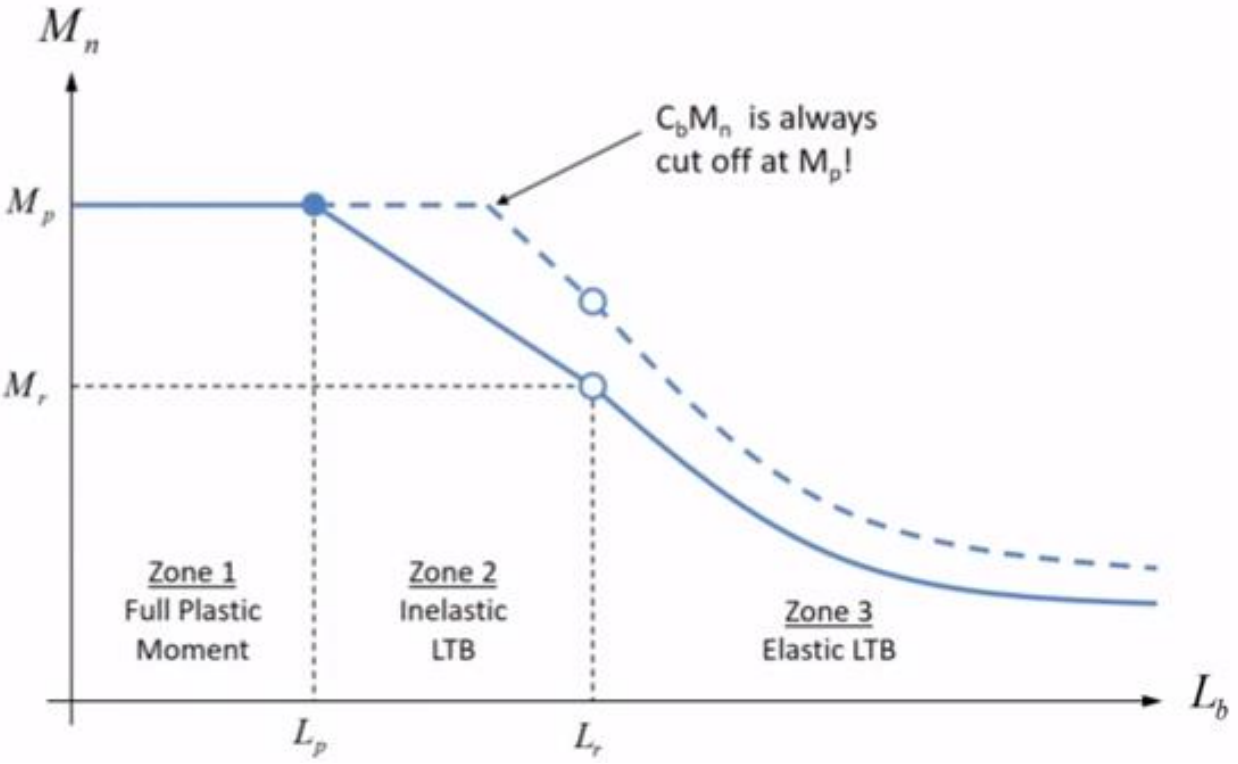
- Plot of M_n vs. L_b for a C_b value of 1.0:

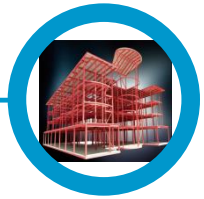




Lateral Torsional Buckling

- However, M_n can't be larger than M_p





Lateral Torsional Buckling

- These zones are split based on two limiting unbraced lengths:

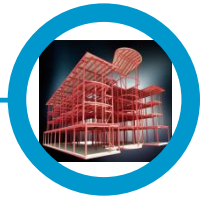
$$L_p = 1.76 r_y \sqrt{\frac{E}{F_y}}$$

Eqn. F2-5, Pg. 16.1-48

$$L_r = 1.95 r_{ts} \frac{E}{0.7F_y} \sqrt{\frac{Jc}{S_x h_o} + \sqrt{\left(\frac{Jc}{S_x h_o}\right)^2 + 6.76\left(\frac{0.7F_y}{E}\right)^2}}$$

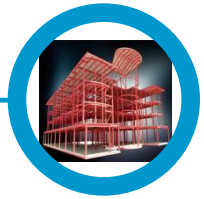
Eqn. F2-6, Pg. 16.1-48

  - c = 1.0 for doubly-symmetric I-shapes
  - r_{ts} = effective radius of gyration for LTB (in)
  - h_o = distance between flange centroids (in)
  - J = torsional constant (in^4)



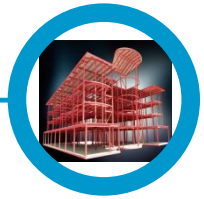
Lateral Torsional Buckling

- As stated, we essentially have three zones:
  - Zone 1: Full plastic moment
    - $L_b \leq L_p$
    - This is the same as before!
  - Zone 2: Inelastic LTB
    - $L_p < L_b \leq L_p$
    - Straight line fit between M_p and $0.7F_yS_x$
  - Zone 3: Elastic LTB
    - $L_b > L_r$
    - Same as the derivation for LTB shown before



Lateral Torsional Buckling

- As stated, we essentially have three zones:
  - Zone 1: Full plastic moment
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    - $L_b > L_r$
    - Same as the derivation for LTB shown before



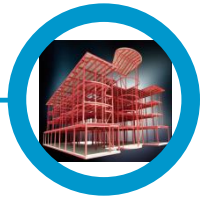
Lateral Torsional Buckling

- Zone 2: Inelastic LTB:

$$M_n = C_b\left[M_p - \left(M_p - 0.7F_yS_x\right)\left(\frac{L_b - L_p}{L_r - L_p}\right)\right] \le M_p$$

Eqn. F2-2, Pg. 16.1-47

  - C_b = moment gradient modifier
  - F_y = yield stress (ksi)
  - S_x = strong-axis section modulus (in^3)
  - L_p, L_r = limiting unbraced lengths (ft)

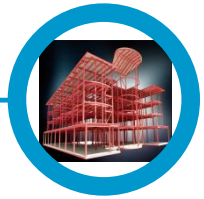


Lateral Torsional Buckling

- Zone 2: Inelastic LTB:
  - Like before, we can write this as:

$$M_n = \min\left\{\begin{array}{c} C_b\left[M_p - \left(M_p - M_r\right)\left(\dfrac{L_b - L_p}{L_r - L_p}\right)\right] \\ M_p \end{array}\right\}$$

  - Note that $0.7F_yS_x$ has been replaced with M_r
    - M_r = yield moment reduced due to residual stresses
    - This notation will be seen again.



Lateral Torsional Buckling

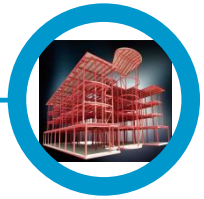
- Those are some pretty lengthy equations
  - Luckily, the Specifications give us a shortcut

Z_x

Table 3-2 (continued)
W-Shapes
Selection by Z_x

F_y = 50 ksi

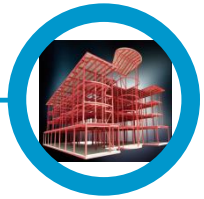
Shape	Z_x	M_{px}/Ω_b	$\phi_b M_{px}$	M_{rx}/Ω_b	$\phi_b M_{rx}$	BF/Ω_b	$\phi_b BF$	L_p	L_r	I_x	V_{nx}/Ω_v	$\phi_v V_{nx}$
		kip-ft	kip-ft	kip-ft	kip-ft	kips	kips				kips	kips
	$in.^3$	ASD	LRFD	ASD	LRFD	ASD	LRFD	ft	ft	$in.^4$	ASD	LRFD
W40×149[f]	598	1490	2240	896	1350	38.3	57.4	8.09	23.6	9800	432	650
W36×150	581	1450	2180	880	1320	34.4	51.9	8.72	25.3	9040	449	673
W27×178	570	1420	2140	882	1330	21.6	32.5	11.5	36.4	7020	403	605
W33×152	559	1390	2100	851	1280	31.7	48.3	8.72	25.7	8160	425	638
W24×192	559	1390	2100	858	1290	18.4	28.0	10.8	39.7	6260	413	620



Lateral Torsional Buckling

- Zone 2: Inelastic LTB:
  - L_p and L_r are conveniently calculated for us.
  - Also, our plastic moment and our reduced yield moment are also listed:
    - $\phi_b M_{px} = \phi_b F_y Z_x$
    - $\phi_b M_{rx} = \phi_b (0.7 F_y Z_x)$
  - There's also another convenience listed
    - If we rearrange the first part of the Zone 2 expression:

$$M_n = C_b\left[M_{px} - \left(M_{px} - M_{rx}\right)\left(\frac{L_b - L_p}{L_r - L_p}\right)\right] = C_b\left[M_{px} - \left(\frac{M_{px} - M_{rx}}{L_r - L_p}\right)\left(L_b - L_p\right)\right] = C_b\left[M_{px} - BF\left(L_b - L_p\right)\right]$$



Lateral Torsional Buckling

- Zone 3: Elastic LTB:

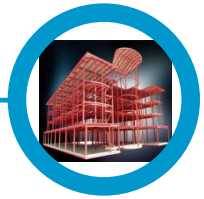
$$M_n = F_{cr} S_x \leq M_p$$

Eqn. F2-3, Pg. 16.1-47

  - F_{cr} = elastic critical LTB stress (ksi)

  – This, of course, can be rewritten as:

$$M_n = \min \begin{Bmatrix} F_{cr} S_x \\ M_p \end{Bmatrix}$$



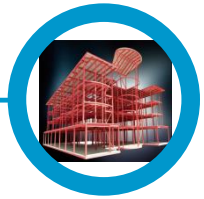
Lateral Torsional Buckling

- Zone 3: Elastic LTB:
  - To compute F_{cr}:

$$F_{cr} = \frac{C_b \pi^2 E}{\left(\frac{L_b}{r_{ts}}\right)^2} \sqrt{1 + 0.078 \frac{Jc}{S_x h_o} \left(\frac{L_b}{r_{ts}}\right)^2}$$

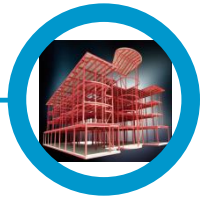
Eqn. F2-4, Pg. 16.1-47

  - Unfortunately, there isn't a design aid to help us calculate this.
    - We just have to work our way through this one.



Lateral Torsional Buckling

- Summary of equations:
  - Zone 1: $$\phi_b M_n = \phi_b M_p = \phi_b F_y Z_x$$
  - Zone 2: $$\phi_b M_n = \min\left\{\begin{matrix} C_b\left[\phi_b M_{px} - \phi_b BF\left(L_b - L_p\right)\right] \\ \phi_b M_{px} \end{matrix}\right\}$$
  - Zone 3: $$\phi_b M_n = \min\left\{\begin{matrix} \phi_b F_{cr} S_x \\ \phi_b M_{px} \end{matrix}\right\}$$



Lateral Torsional Buckling

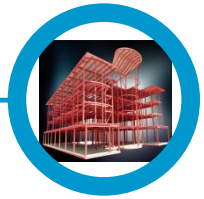
- Summary of equations (cont'd):
  - C_b factor:
    - Also found in *Tab. 3-1*

$$C_b = \frac{12.5M_{max}}{2.5M_{max} + 3M_A + 4M_B + 3M_C}$$

  - F_{cr} stress:

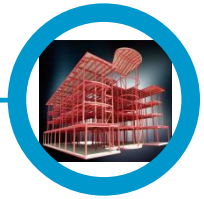
$$F_{cr} = \frac{C_b \pi^2 E}{\left(\frac{L_b}{r_{ts}}\right)^2} \sqrt{1 + 0.078 \frac{Jc}{S_x h_o} \left(\frac{L_b}{r_{ts}}\right)^2}$$

  - L_p, L_r, and $\phi_b BF$ can be found in *Tab. 3-2*



Lateral Torsional Buckling

- A W24×76 is to be used as a simply supported beam to resist a concentrated load at midspan.
  - The beam will be laterally braced at the supports and at midspan
  - Assuming that the self weight of the beam may be neglected, determine the flexural capacity of the beam for the following cases:
    - Case 1: L = 10 feet
    - Case 2: L = 30 feet
    - Case 3: L = 50 feet



Lateral Torsional Buckling

- beam parameters (W24x76):

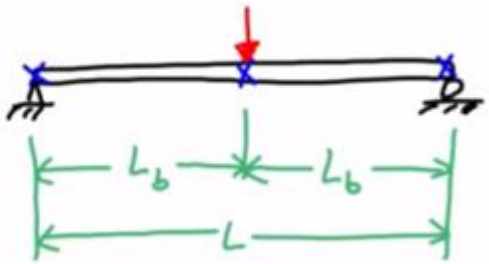
- Tab. 1-1: $S_x = 176 in^3$, $r_{ts} = 2.33 in$

  $h_o = 23.2 in$, $J = 2.68 in^4$

- Tab. 3-2: $\phi_b M_{px} = 750$ ft-kip , $\phi_b M_{rx} = 462$ ft-kip

  $\phi_b BF = 22.6$ kip , $L_p = 6.78$ ft , $L_r = 19.5$ ft

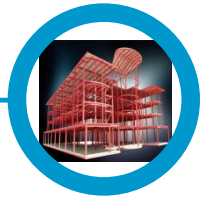
- Tab. 3-1:



$L_b = L/2$

$C_b = 1.67$

$L_p = 6.78$ ft , $L_r = 19.5$ ft



Lateral Torsional Buckling

Case 1: $L = 10\,ft \;\therefore\; L_b = 5\,ft \;\therefore$ **Zone 1**

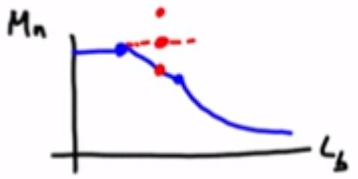
$$\therefore \boxed{\phi_b M_n = \phi_b M_{px} = 750\ ft\text{-}kip} \quad \text{ANS (Case 1)}$$

Case 2: $L = 30\,ft \;\therefore\; L_b = 15\,ft \;\therefore$ Zone 2

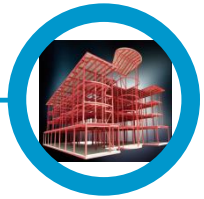
$$\therefore \phi_b M_n = \min \left\{ \begin{array}{c} C_b \left[\phi_b M_{px} - \phi_b BF \left(L_b - L_p \right) \right] \\ \phi_b M_{px} \end{array} \right\}$$

$$= \min \left\{ \begin{array}{c} 1.67 \left[750\ ft\text{-}kip - (27.6\ kip)(15\,ft - 6.78\,ft) \right] \\ 750\ ft\text{-}kip \end{array} \right\}$$

$$= \min \left\{ 1.67 \left[564.2\ ft\text{-}kip \right] \right\} = \min \left\{ \begin{array}{c} 942.3\ ft\text{-}kip \\ 750\ ft\text{-}kip \end{array} \right\}$$



$\phi_b M_n = 750\ ft\text{-}k$



Lateral Torsional Buckling

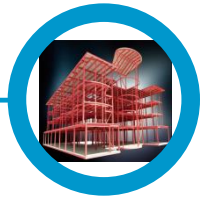
- Case 3: $L = 50\ ft \therefore L_b = 25\ ft \therefore$ Zone 3

$$\therefore F_{cr} = \frac{C_b \pi^2 E}{\left(L_b/r_{ts}\right)^2}\sqrt{1 + 0.078\left(\frac{J_c}{S_x h_o}\right)\left(L_b/r_{ts}\right)^2}$$

$$= \frac{1.67(\pi^2)(28000\ ksi)}{\left(\frac{300"}{2.33"}\right)^2}\sqrt{1 + 0.078\left[\frac{(2.68\ in^4)(1.0)}{(176\ in^3)(23.2")}\right]\left(\frac{300"}{2.33"}\right)^2} = 39.20\ ksi$$

$$\therefore \phi_b M_n = \min\left\{\begin{matrix}\phi_b F_{cr} S_x \\ \phi_b M_{px}\end{matrix}\right\} = \min\left\{\begin{matrix}0.90(39.20\ ksi)(176\ in^3)\left(1\ ft/12\ in\right) \\ 750\ ft\text{-}kip\end{matrix}\right\}$$

$$= \min\left\{\begin{matrix}575.0\ ft\text{-}kip \\ 750\ ft\text{-}kip\end{matrix}\right\} \qquad \phi_b M_n = 575.0\ ft\text{-}kip$$



Lateral Torsional Buckling

- Case 3: $L = 50\ ft \therefore L_b = 25\ ft \therefore$ Zone 3

$$\therefore F_{cr} = \frac{C_b \pi^2 E}{\left(L_b/r_{ts}\right)^2}\sqrt{1 + 0.078\left(\frac{J_c}{S_x h_o}\right)\left(L_b/r_{ts}\right)^2}$$

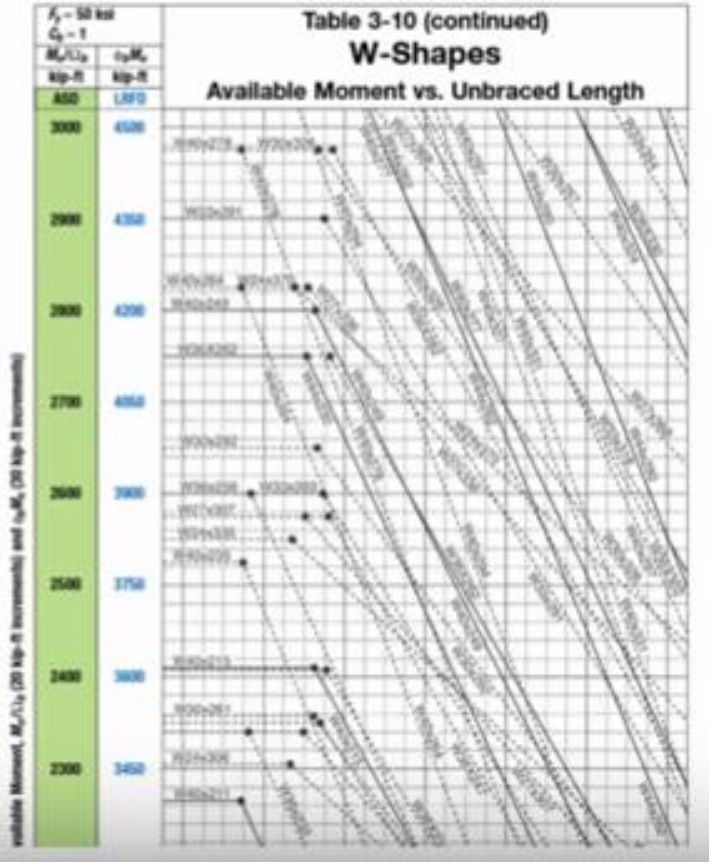
$$= \frac{1.67(\pi^2)(28000\ ksi)}{\left(\frac{300"}{2.33"}\right)^2}\sqrt{1 + 0.078\left[\frac{(2.68\ in^4)(1.0)}{(176\ in^3)(23.2")}\right]\left(\frac{300"}{2.33"}\right)^2} = 39.20\ ksi$$

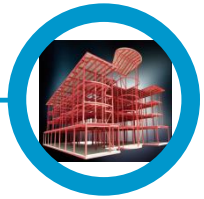
$$\therefore \phi_b M_n = \min\left\{\begin{matrix}\phi_b F_{cr} S_x \\ \phi_b M_{px}\end{matrix}\right\} = \min\left\{\begin{matrix}0.90(39.20\ ksi)(176\ in^3)\left(1\ ft/12\ in\right) \\ 750\ ft\text{-}kip\end{matrix}\right\}$$

$$= \min\left\{\begin{matrix}575.0\ ft\text{-}kip \\ 750\ ft\text{-}kip\end{matrix}\right\} \qquad \phi_b M_n = 575.0\ ft\text{-}kip$$

Lateral Torsional Buckling

- Note that some of the lines are dashed and some are solid.
- The solid lines refer to the most economical section in a region.
  - Much like the bold values in the Z_x tables.

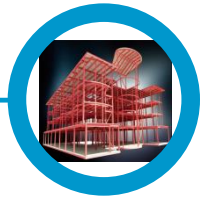




Lateral Torsional Buckling

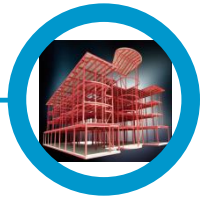
- Tab. 3-10 plots the LTB curve for cases where $C_b = 1.0$
- Since we rarely deal with this when designing beams, we must divide the moment by C_b to use the design aid.
  - Therefore, to design a beam, we locate a section with the following two values:

Enter Tab. 3-10 With:
L_b
M_u / C_b



Lateral Torsional Buckling

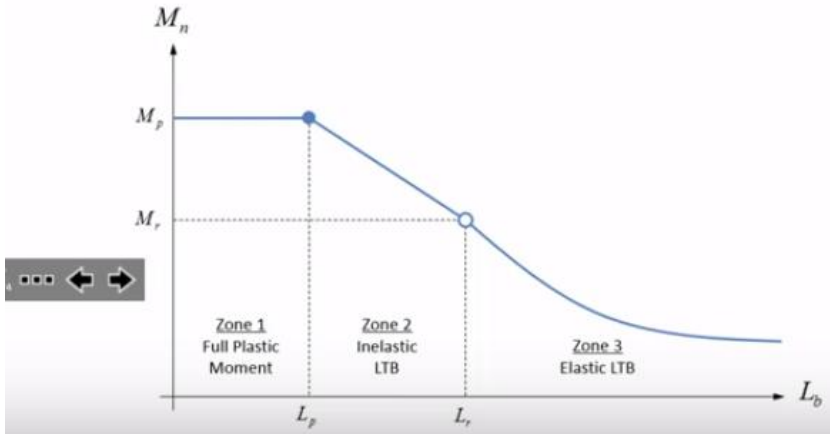
- To design a beam that is discretely braced, we need a means to select beams based on LTB.
- This can be difficult since each beam has its own LTB curve.

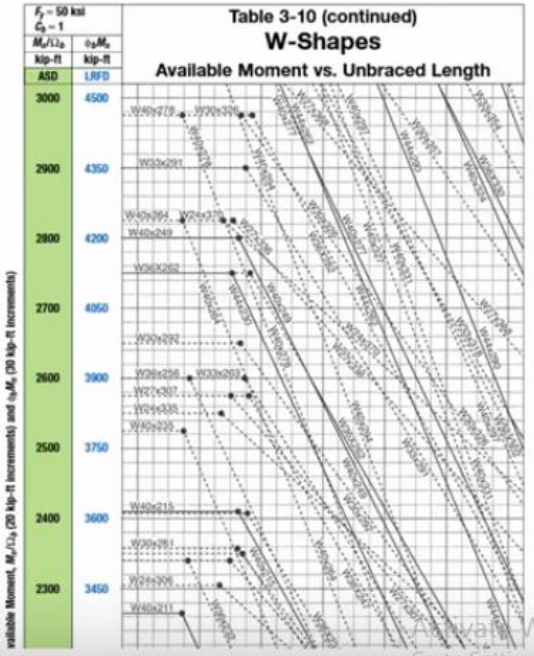


Lateral Torsional Buckling

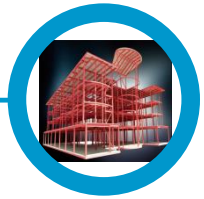
Pg. 3-99 ★

- *Tab. 3-10:*
  - This may look messy, but its nothing more than the LTB curve plotted for every beam.



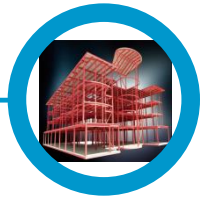


Activate Windows
Go to Settings to activate Windows



Lateral Torsional Buckling

- Note that some of the lines are dashed and some are solid.
- The solid lines refer to the most economical section in a region.
  - Much like the bold values in the Z_x tables.



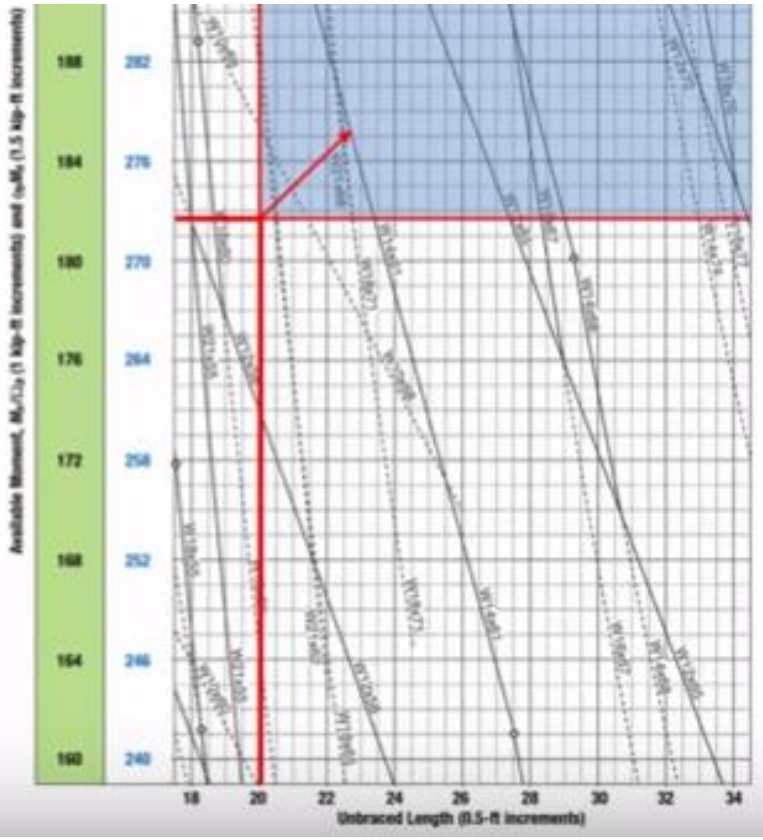
Lateral Torsional Buckling

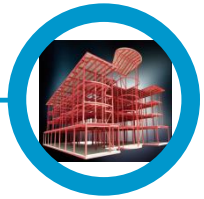
- Tab. 3-10 plots the LTB curve for cases where $C_b = 1.0$
- Since we rarely deal with this when designing beams, we must divide the moment by C_b to use the design aid.
  - Therefore, to design a beam, we locate a section with the following two values:

Enter Tab. 3-10 With:
L_b
M_u / C_b

Lateral Torsional Buckling

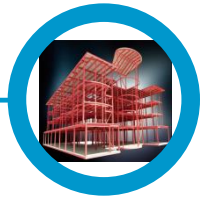
- To design a beam, simply locate the appropriate M_u/C_b and L_b value in the chart.
  - For example:
    - M_u/C_b = 272.5 ft-kip
    - L_b = 20 ft
  - Anything in the blue shaded region is safe!





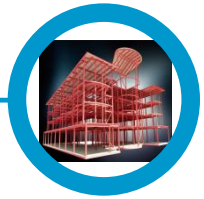
Lateral Torsional Buckling

- Note:
  - Once we select a section, we **must** check its capacity.
    - Recall that a beam's capacity cannot exceed M_p.
  - Therefore, it is possible that, because we enter the chart with M_u/C_b, we may select a section that ends up not being valid!
    - If that happens, we simply have to iterate and select the next lightest section.



Lateral Torsional Buckling

- Select the lightest W-section to support a uniformly distributed service dead load of 600 pounds per foot (including the self-weight of the beam) and a concentrated service live load of 40 kips at midspan.
  - The beam is 20 feet long and simply supported.
  - In addition to flexural requirements, ensure that all beams have adequate stiffness to resist a service live load deflection of L/300.
- For this example, we will design four different beams!
  - Case 1: L_b = 20', C_b = 1.0
  - Case 2: L_b = 20', Compute C_b
  - Case 3: L_b = 10', C_b = 1.0
  - Case 4: L_b = 10', Compute C_b
- Doing this, we'll be able to see the effect of L_b and C_b has on the final beam selection



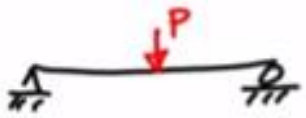
Lateral Torsional Buckling

- beam parameters: $w_D = 0.6\ kip/ft$, $w_L = 0.72\ kip/ft$

$P_L = 40\ kip$, $P_D = 64\ kip$

$L = 20\ ft = 240''$

- serviceability requirements: $\delta_{allow} = \frac{L}{300}$, $\delta_{LL} = \frac{P_L L^3}{48 E I_x}$ (Tab. 3-23, Case 7)



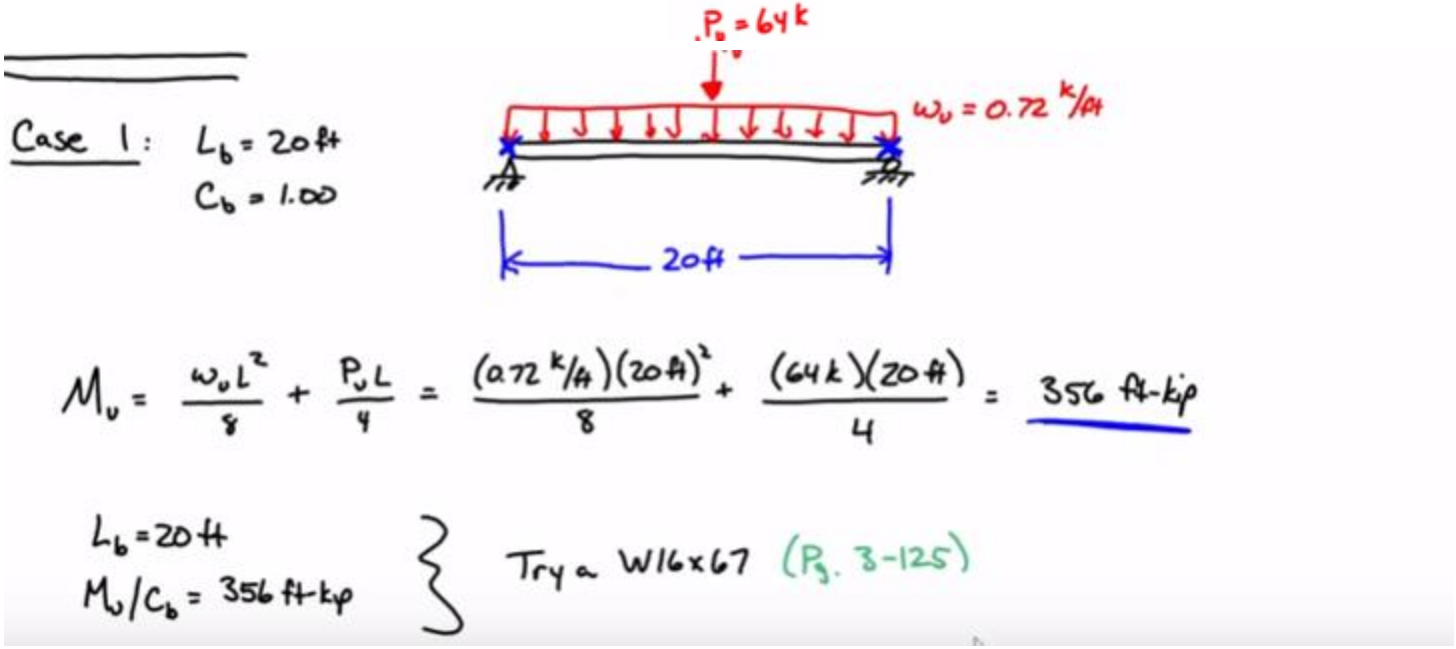
$$\delta_{LL} \le \delta_{allow}$$

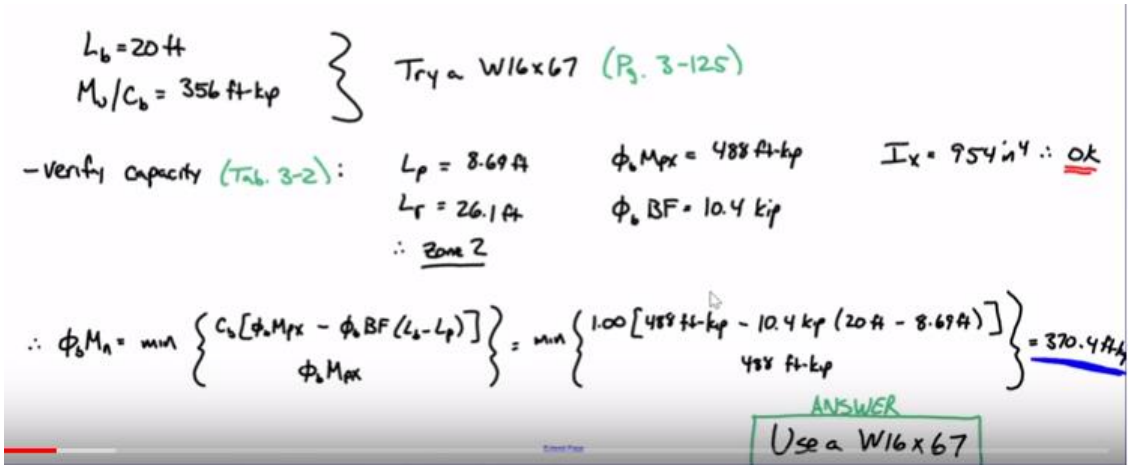
$$\frac{P_L L^3}{48 E I_x} \le \frac{L}{300}$$

$$\frac{P_L L^3}{48 E}\left(\frac{300}{L}\right) \le I_x$$

$$\therefore I_{x_{min}} = \frac{300 P_L L^2}{48 E} = \frac{300\,(40\ kip)(240'')^2}{48\,(29000\ ksi)} = 497\ in^4$$

Lateral Torsional Buckling



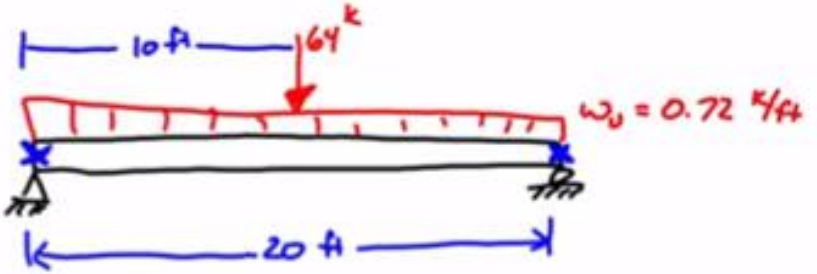


Lateral Torsional Buckling

Ex. 23 (cont'd)

Case 2: $L_b = 20$ ft

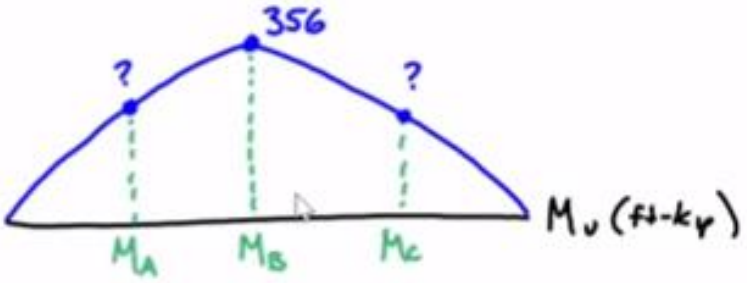
compute C_b

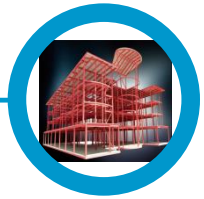


Tab. 3-23

① $M_u(x) = \frac{w_u x}{2}(L-x)$

⑦ $M_u(x) = \frac{P_u x}{2}$





Lateral Torsional Buckling

$$\therefore \quad M_A = M_c = \frac{w_u}{2}(5\,ft)(L - 5\,ft) + \frac{P_u}{2}(5\,ft)$$

$$= \frac{(0.72\,k/ft)}{2}(5\,ft)(20\,ft - 5\,ft) + \frac{64\,kip}{ft}(5\,ft)$$

$$= 187\,ft\text{-}kip$$

$M_B = M_{max} = 356\,ft\text{-}kip$

$$\therefore \quad C_b = \frac{12.5\,M_{max}}{2.5M_{max} + 3M_A + 4M_B + 3M_c} = \frac{12.5(356)}{2.5(356) + 3(187) + 4(356) + 3(187)} = 1.298$$

$\therefore \quad L_b = 20\,ft$

$M_u/C_b = \dfrac{356\,ft\text{-}kip}{1.298} = 274.3\,ft\text{-}kip$

Try a W14 x 61

- analysis: (Tab. 3-2)

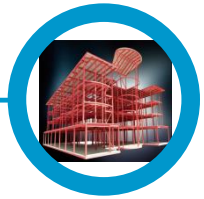
$L_p = 6.65\,ft$

$L_r = 27.5\,ft$

$\therefore$ Zone 2

$\phi_b M_{px} = 383\,ft\text{-}kip$

$\phi_b BF = 7.48\,kip$



Lateral Torsional Buckling

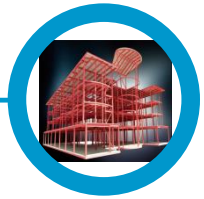
$$\therefore \phi_b M_n = \min \begin{Bmatrix} C_b[\phi_b M_{px} - \phi_b BF(L_b - L_p)] \\ \phi_b M_{px} \end{Bmatrix}$$

$$= \min \begin{Bmatrix} 1.298[383\ \text{ft-kip} - 7.48\ \text{ft-kip}(20\text{ft} - 8.65\text{ft})] \\ 383\ \text{ft-kip} \end{Bmatrix} = \min \begin{Bmatrix} 386.9\ \text{ft} \\ 383\ \text{ft-k} \end{Bmatrix}$$

$$= 383\ \text{ft-kip}$$

ANS (Case 2)

Use a W14x61



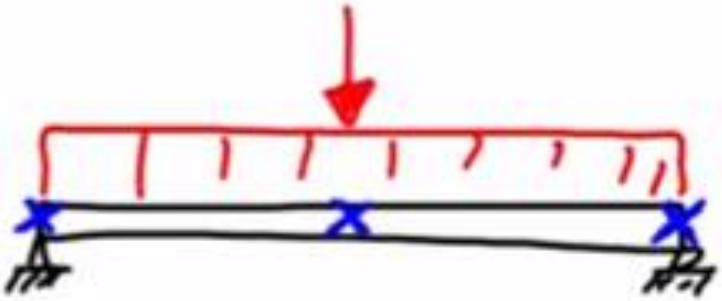
Lateral Torsional Buckling

$M_u = 356$ ft-kip

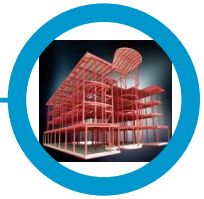
$C_b = 1.0$

$\therefore M_u/C_b = 356$ ft-kip

$L_b = 10$ ft

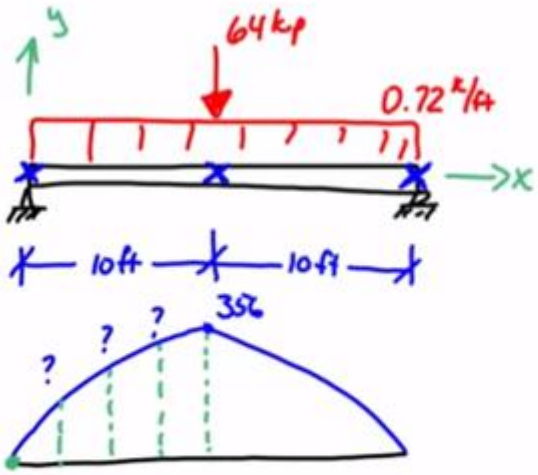


Use a W18x55



Lateral Torsional Buckling

Case 4:



$M(x) = \frac{wx}{2}(L-x)$ [Dist. Load]

$M(x) = \frac{Px}{2}$ [Point Load]

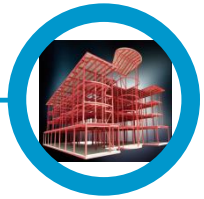
X (ft)	M_{dist}	M_{point}	M_{total}	
0	0	0	0	
2.5	15.75	80	95.75	M_A
5	27	160	187	M_B
7.5	33.75	240	273.75	M_C
10	36	320	356	M_{max}

$\therefore M_u/C_b = \frac{356 \text{ ft-kip}}{1.620} = 219.8 \text{ ft-kip}$

$\therefore C_b = 1.620$

$L_b = 10$ ft

W16 x 40 ($\phi_b M_{px} = 274$ ft-kip) NG



Lateral Torsional Buckling

W21x44 : $L_p = 4.45$ ft, $L_r = 13.0$ ft (Tab 3-2) $\therefore$ Zone 2

$\phi_b M_{px} = 358$ ft-kip

$\phi_b BF = 16.8$ kip

$I_x = 843$ in^4 $\therefore$ OK

$$\therefore \phi_b M_n = \min \begin{Bmatrix} C_b\left[\phi_b M_{px} - \phi_b BF\,(L_b - L_p)\right] \\ \phi_b M_{px} \end{Bmatrix} = \min \begin{Bmatrix} 1.620\left[358 \text{ ft-kip} - 16.8 \text{ kip}\,(10\text{ft} - 4.45\text{ft})\right] \\ 358 \text{ ft-kip} \end{Bmatrix}$$

$$= \min \begin{Bmatrix} 428.9 \text{ ft-kip} \\ 358 \text{ ft-kip} \end{Bmatrix} = 358 \text{ ft-kip}$$

$\phi M_n = 358$ ft-kip

$M_u = 356$ ft-kip

ANS (C)

$\therefore$ Use W21x44


Questions?